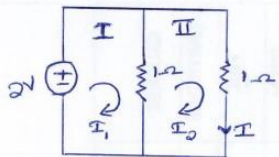
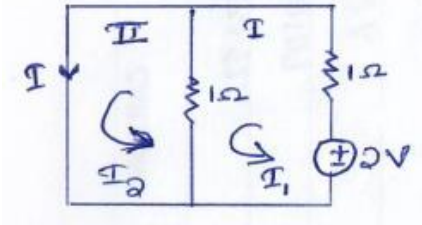
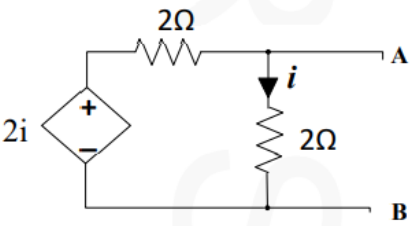
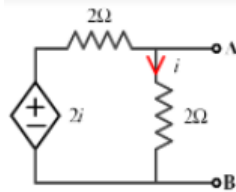


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
THIRD SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021		
Course Code: EET201		
Course Name: CIRCUITS AND NETWORKS		
PART A		
		<i>Answer all questions. Each question carries 3 marks</i>
	Q1	State and explain reciprocity theorem using an example.
	Ans	In a linear, bilateral, active, single source network, the ratio of excitation to response is a constant when their positions are interchanged.  Applying mesh in I $2 - 1(I_1 - I_2) = 0$ $-I_1 + I_2 = -2$ $I_1 - I_2 = 2 \text{ ----- (1)}$ Applying mesh in II $-1(I_2 - I_1) - I_2 = 0 \text{ ----- (2)}$ $-I_2 + I_1 - I_2 = 0$ $I_1 - 2I_2 = 0 \text{ ----- (3)}$ Solving, $I_1 - I_2 = 2$ $I_1 - 2I_2 = 0$ $I_1 = 4 \quad I_2 = 2$ $\therefore I = I_2 = 2$ $\therefore E/I = 2/2 = 1$

		 Mesh I, $2 - 1(I_1) - 1(I_1 - I_2) = 0$ $2 - I_1 - I_1 + I_2 = 0$ $2 - 2I_1 + I_2 = 0$ $2I_1 - I_2 = 2 \text{ ----- (1)}$ Mesh II $-1(I_2 - I_1)$ $-I_2 + I_1 = 0$ $I_1 - I_2 = 0 \text{ ----- (2)}$ $I_1 = 2$ $I_2 = 2$ $I_2 = I = 2$ $\therefore E/I = 2/2 = 1$ Hence reciprocity theorem is verified.
Q2		Determine the Norton's equivalent circuit of the following network. 
Ans		<u>Norton's Equivalent Circuit:</u> When 'A' and 'B' made short circuited, $i=0$ Thus, $I_{AB} = I_N = 0A$



To find the Norton's equivalent resistance, connect an arbitrary source V_{dc} across 'A' and 'B'.

Applying KVL to loop 'ECDFE'

$$2i + 2(I_{dc} - i) - 2i = 0 \quad \longrightarrow \quad i = I_{dc}$$

Applying KVL to loop 'ACDBA'

$$V_{dc} - 2i = 0$$

$$V_{dc} = 2i = 2I_{dc}$$

$$\therefore \frac{V_{dc}}{I_{dc}} = Z_{AB} = Z_N = 2\Omega$$

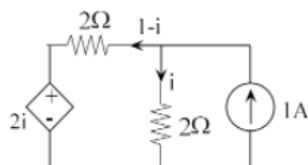
OR

Connect an arbitrary current source of 1A across A and B and find the voltage across that current source.

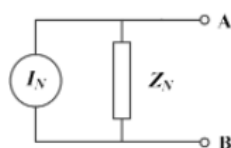
$$2(1-i) + 2i - 2i = 0$$

Hence $i=1A$; voltage across 1A source = 2V

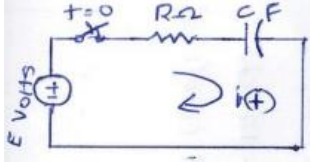
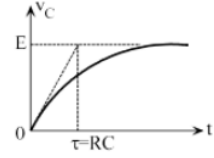
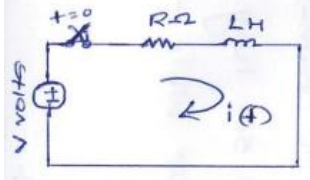
$$Z_N = 2V/1A = 2\Omega$$



Norton's equivalent circuit:



Q3	Define time constant of a circuit. Illustrate and explain does time constant affect the charging time of the capacitor in series RC circuits connected to a DC source?
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Ans	 $Ri(t) + \frac{1}{C} \int_0^{\infty} i(t) dt = E$ $i(t) = \frac{E}{R} e^{-\left(\frac{1}{RC}\right)t} = \frac{E}{R} e^{-\left(\frac{1}{\tau}\right)t} \quad \text{where, } \tau = RC = \text{time constant}$ $v_C(t) = \left(1 - e^{-\left(\frac{1}{\tau}\right)t}\right) E$  As the time constant increases, charging time required to reach the steady voltage across capacitor increases or speed of response decreases.
Q4	Derive the expression for the current in a series RL circuit when connected to a DC source of voltage V, at time t = 0. Assume zero initial conditions.
Ans	 $Ri(t) + L \frac{d}{dt} i(t) = V$ $RJ(s) + LsI(s) = \frac{V}{s}$ $I(s) = \frac{\left(\frac{V}{R}\right)}{s} + \frac{\left(-\frac{V}{R}\right)}{\left(s + \frac{R}{L}\right)}$ $i(t) = \frac{V}{R} \left(1 - e^{-\left(\frac{Rt}{L}\right)}\right)$
Q5	Obtain the transfer function of a typical series RLC circuit. Take the voltage across the capacitor as the output variable.
Ans	Assume zero initial conditions.

		$TF = \frac{LT[V_{out}]}{LT[V_{in}]} = \frac{LT\left[\frac{1}{C}\int_0^\infty i(t)dt\right]}{LT\left[Ri(t) + L\frac{d}{dt}i(t) + \frac{1}{C}\int_0^\infty i(t)dt\right]} = \frac{\left[\frac{1}{sC}\right]I(s)}{\left[R + sL + \frac{1}{sC}\right]I(s)}$ $= \frac{\left[\frac{1}{sC}\right]}{\left[\frac{s^2LC + sRC + 1}{sC}\right]} = \frac{1}{LCs^2 + RCs + 1}$																												
	Q6	Define coefficient of coupling in coupled circuits. What are its maximum and minimum values?																												
	Ans	The coefficient of coupling can be defined as fraction of magnetic flux produced by current in one coil that links with the other coil. It is represented as K. Max. value is 1 & Min. value is 0 $K = M/\sqrt{L_1 L_2}$																												
	Q7	A series RLC circuit with $R = 10\Omega$, $L = 2H$ and $C = 0.5F$ is connected to a 230V, variable frequency AC source. Determine the frequency of the source at which the circuit current is maximum. Also find the maximum current.																												
	Ans	$f_r = 1/2\pi\sqrt{LC}$ $f_r = 1/2\pi\sqrt{2 \times 0.5}$ $f_r = 0.159$ Max. current at resonance, $I_{max} = V_s/R$ $I = 230/10$ $I = 23 \text{ A}$																												
	Q8	Describe the variation of the impedance, power factor and current as a function of frequency in a series resonant circuit.																												
	Ans	<table><tr><th>$f(Hz)$</th><th>$X_L(\Omega)$</th><th>$X_C(\Omega)$</th><th>$Z(\Omega)$</th><th>Phase angle</th><th>pf</th><th>current</th></tr><tr><td>0</td><td>0</td><td>∞</td><td>∞</td><td>-90°</td><td>leading</td><td>0</td></tr><tr><td>f_r</td><td>$\omega_r L$</td><td>$1/\omega_r C$</td><td>R</td><td>0°</td><td>unity</td><td>maximum</td></tr><tr><td>∞</td><td>∞</td><td>0</td><td>∞</td><td>$+90^\circ$</td><td>lagging</td><td>0</td></tr></table> Impedence Power factor Current	$f(Hz)$	$X_L(\Omega)$	$X_C(\Omega)$	$Z(\Omega)$	Phase angle	pf	current	0	0	∞	∞	-90°	leading	0	f_r	$\omega_r L$	$1/\omega_r C$	R	0°	unity	maximum	∞	∞	0	∞	$+90^\circ$	lagging	0
$f(Hz)$	$X_L(\Omega)$	$X_C(\Omega)$	$Z(\Omega)$	Phase angle	pf	current																								
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f_r	$\omega_r L$	$1/\omega_r C$	R	0°	unity	maximum																								
∞	∞	0	∞	$+90^\circ$	lagging	0																								

Q9	What are h-parameters? Why are they called hybrid parameters?	
Ans	$(V_1, V_2) = f(I_1, V_2)$ $V_1 = h_{11} I_1 + h_{12} V_2$ -----① $I_2 = h_{21} I_1 + h_{22} V_2$ -----② $\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$ Out of four parameters, one is measured in Ohm, one in mho and other two are dimension less. Since these parameters have mixed dimensions, so they are called hybrid parameters.	
Q10	Show that the overall T-parameter matrix of two cascaded 2-port networks is the product of the T-parameter matrix of the individual networks.	
Ans	$(V_1, I_1) = f(\dots)$ $V_{1X} = A_X V_{2X} - B_X I_{2X}$ ----- ① $V_{2X} = C_X V_{2Y} - D_X I_{2Y}$ ----- ② $V_{1Y} = A_Y V_{2Y} - B_Y I_{2Y}$ ----- ③ $V_{2Y} = C_Y V_{2Y} - D_Y I_{2Y}$ ----- ④ $V_1 = V_{1X} \quad V_2 = V_{2Y} \quad V_{2X} = V_{1Y}$ $I_1 = I_{1X} \quad I_2 = I_{2Y} \quad I_{2X} = -I_{1Y}$ $I_{2X} = -I_{1Y}$ ① & ② in matrix form is, $\begin{bmatrix} V_{1X} \\ I_{1X} \end{bmatrix} = \begin{bmatrix} A_X & B_X \\ C_X & D_X \end{bmatrix} \begin{bmatrix} V_{2X} \\ -I_{2X} \end{bmatrix}$ ③ & ④ in matrix form is, $\begin{bmatrix} V_{1Y} \\ I_{1Y} \end{bmatrix} = \begin{bmatrix} A_Y & B_Y \\ C_Y & D_Y \end{bmatrix} \begin{bmatrix} V_{2Y} \\ I_{2Y} \end{bmatrix}$ Substitute the conditions in the 1 st matrix	

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \begin{bmatrix} V_{1y} \\ I_{1y} \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \begin{bmatrix} A_y & B_y \\ C_y & D_y \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$$

Where,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_x & B_x \\ C_x & D_x \end{bmatrix} \begin{bmatrix} A_y & B_y \\ C_y & D_y \end{bmatrix}$$

PART B

Answer any one full question from each module. Each question carries 14 marks

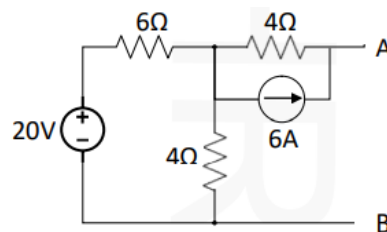
Module 1

Q11

For the network given below,

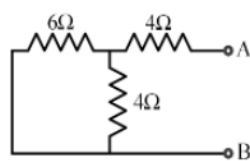
a) Find the Thevenin's equivalent circuit across the terminals A and B.

b) Determine the power dissipated in a 10Ω resistance when it is connected across the terminals A and B.



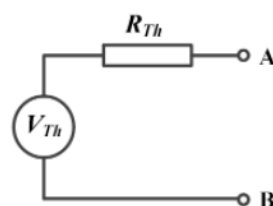
Ans

a) **Thevenin's equivalent circuit:**



$$R_{Th} = R_{AB} = 4 + (4 \parallel 6) = 4 + \frac{4 \times 6}{4 + 6} = 4 + 2.4 = 6.4\Omega$$

$$V_{Th} = V_{AB} = 32V$$

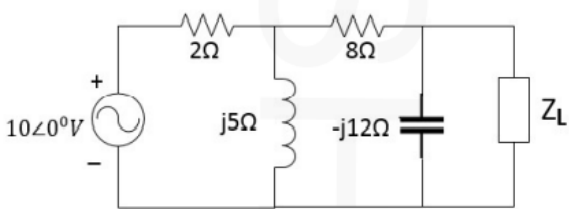


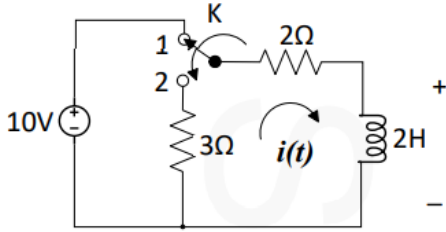
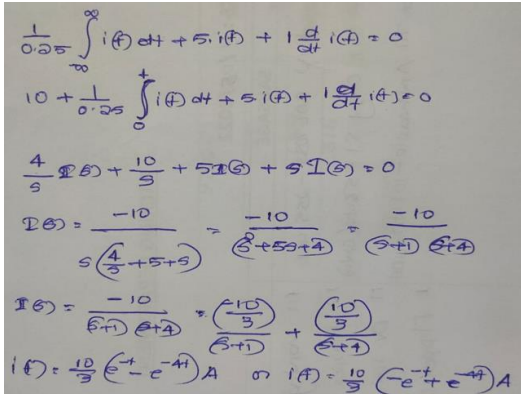
Thevenin's equivalent circuit

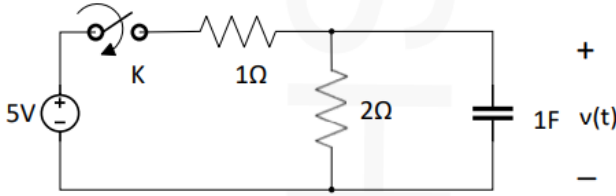
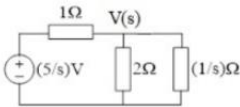
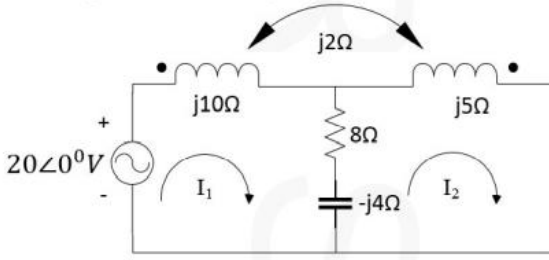
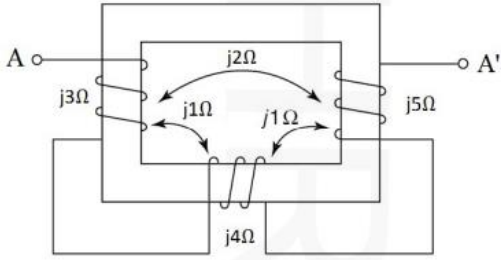
$$R_{Th} = 6.4\Omega$$

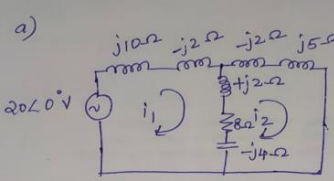
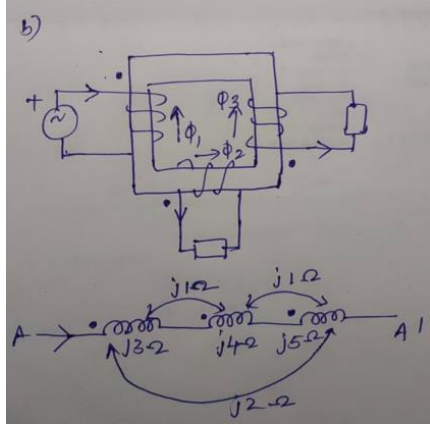
$$V_{Th} = 32V$$

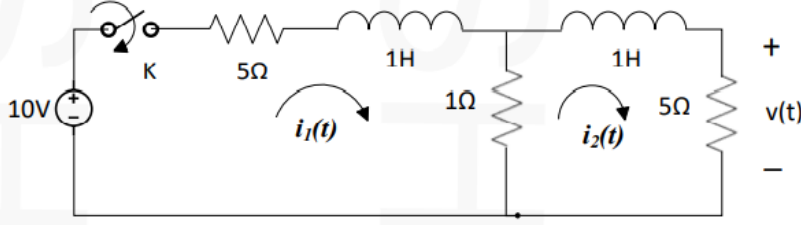
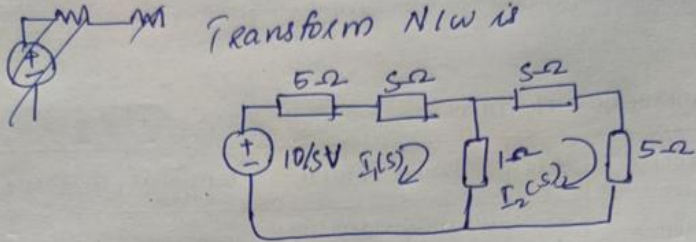
b) Power dissipated in 10Ω resistor connected across A and B,

		$P_{10\Omega} = I^2 \cdot R = \left[\frac{32}{(6.4+10)} \right]^2 \times 10 = 38.073W$ Or $P_{10\Omega} = \frac{V_{10\Omega}^2}{R} = \frac{\left[\frac{32 \times 10}{(6.4+10)} \right]^2}{10} = 38.073W$
	Q12	In the circuit given below, a) Determine the value of the load impedance for maximum power transferred by the source to the load. b) Find the maximum power transferred. 
	Ans	a) Load impedance for maximum power transfer:- $Z_{Th} = Z_{Load} = -j12 \parallel [8 + (j5 \parallel 2)]$ $= 7.843 \angle -36.631^\circ \Omega = 6.294 - j4.68\Omega$ Load impedance for maximum power transfer, $Z_L = Z_{Th}^* = 6.294 + j4.68\Omega$ b) Maximum power transfer:- $V_{Th} = V_{Load} = V_{-j12\Omega} = 7.47 \angle -18.89^\circ V$ Maximum power, $P_L = \frac{V_{Th}^2}{4R_L} = \frac{7.47^2}{4 \times 6.294} = 2.216W$
Module 2		
	Q13	a) In the circuit shown below, the switch was initially at position 1 and the steady state condition is reached. At $t = 0$, the switch is changed to position 2. Determine the expression for the current $i(t)$, for $t > 0$.

	 b) A 0.25F capacitor with an initial voltage of 10V is connected across a coil of 5Ω resistance and 1H inductance, at time $t = 0$. Determine the expression for the current through the coil for $t > 0$.
Ans	a) Transient Current:- At position-1: steady state condition, inductor acts as SC. $I_0 = 10/2 = 5A$ At position-2: The transient current in the circuit can be found by applying KVL to the closed loop. $3i(t) + 2i(t) + 2 \frac{d}{dt} i(t) = 0$ $5I(s) + 2[sI(s) - I_0] = 0$ $\therefore I(s) = \frac{10}{(2s+5)} = \frac{5}{(s+2.5)}$ $i(t) = 5e^{-2.5t}$ b) The transient current in the circuit can be found by applying KVL to the closed loop. 
Q14	In the circuit given below, the switch K is closed at $t = 0$. a) Determine the expression for the voltage across the capacitor, $v(t)$ for $t > 0$. b) Calculate the value of $v(t)$ at $t = 1$ seconds and its final steady state value.

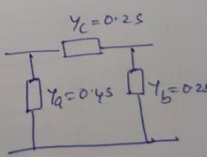
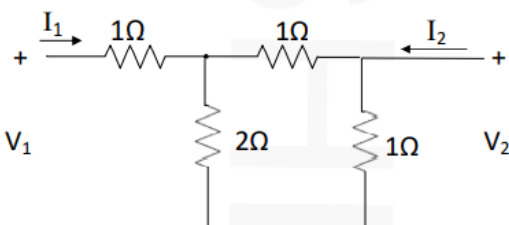
		
Ans	Voltage across capacitor:-	 Using Node analysis, $V(s) - \frac{5}{s} + \frac{V(s)}{2} + sV(s) = 0$ $V(s) = \frac{5}{s(s+1.5)} = \frac{10/3}{s} - \frac{10/3}{s+1.5}$ $v(t) = \frac{10}{3}(1 - e^{-1.5t})$ $v(t=0) = \frac{10}{3}(1 - e^{-1.5 \cdot 0}) = 0.59V$ $v(t=\infty) = \frac{10}{3}(1 - e^{-\infty}) = \frac{10}{3} = 3.33V$
Module 3		
Q15	a) Obtain the conductively coupled equivalent circuit for the network given below. Also write the mesh equations for the equivalent circuit.  b) If the network given below is connected across a 230V, 50Hz AC source, determine the current supplied by the source. 	

Ans	a)  mesh equations:- $\begin{bmatrix} (8+j10-j4) & -(8+j2-j4) \\ -(8+j2-j4) & (-j4+8+j5) \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 20\angle 0^\circ \\ 0 \end{bmatrix}$ $\begin{bmatrix} (8+j6) & -(8-j2) \\ -(8-j2) & (8+j1) \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 20\angle 0^\circ \\ 0 \end{bmatrix}$ b)  $X_{eq} = X_{L1} + X_{L2} + X_{L3} + 2X_{M12} + 2X_{M23} + 2X_{M31}$ $= j3 + j4 + j5 + 2(j1) + 2(j1) + 2(j2) = j20\Omega$ $\text{Current } I = \frac{\text{Voltage}}{X_{eq}} = \frac{230\angle 0^\circ}{20\angle 90^\circ} = 11.5\angle -90^\circ A = -j11.5A$
Q16	The switch K in the circuit given below is closed at $t = 0$. a) Determine the transformed circuit for $t > 0$. Assume zero initial conditions. b) Find the time domain expression for the voltage $v(t)$ across the 5Ω resistor for $t > 0$. Use mesh analysis.

	
Ans	Transform N1W is  Mesh equations $\begin{bmatrix} (6+s) & -1 \\ -1 & 6+s \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} 10/s \\ 0 \end{bmatrix}$ $I_2(s) = \frac{\Delta Z_2}{\Delta Z} \quad \Delta Z = \begin{vmatrix} 6+s & -1 \\ -1 & 6+s \end{vmatrix} = (6+s)^2 - 1 = s^2 + 12s + 35 = (s+5)(s+7)$ $\Delta Z_2 = \begin{vmatrix} 6+s & 10/s \\ -1 & 0 \end{vmatrix} = 0 - \frac{10}{s} = -\frac{10}{s}$ $I_2(s) = \frac{-10}{s(s^2 + 12s + 35)} = \frac{-10}{s(s+5)(s+7)}$ $I_2(s) = \frac{2/7}{s} + \frac{-1}{s+5} + \frac{5/7}{s+7}$ $i_2(t) = \frac{2}{7} - e^{-5t} + \frac{5}{7}e^{-7t} \text{ A}$ $V_{5\Omega}(t) = 5 \times i_2(t) = \frac{10}{7} - 5e^{-5t} + \frac{25}{7}e^{-7t} \text{ V}$
Module 4	
Q17	A 3-phase, 3-wire, 240V system supplies a delta-connected load in which $Z_{AB} = 25\angle 90^\circ \Omega$, $Z_{BC} = 15\angle 30^\circ \Omega$ and $Z_{CA} = 20\angle 0^\circ \Omega$. Determine the phase currents, line currents and total power consumed by the load.

Ans	ABC Sequence $V_{AB} = 240 \angle 0^\circ \text{ V} \quad V_{BC} = 240 \angle -120^\circ \text{ V} \quad V_{CA} = 240 \angle 120^\circ \text{ V}$ $I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{240 \angle 0^\circ}{25 \angle 90^\circ} = 9.6 \angle -90^\circ \text{ A}$ $I_{BC} = \frac{V_{BC}}{Z_{BC}} = \frac{240 \angle -120^\circ}{15 \angle 30^\circ} = 16 \angle -150^\circ \text{ A}$ $I_{CA} = \frac{V_{CA}}{Z_{CA}} = \frac{240 \angle 120^\circ}{20 \angle 0^\circ} = 12 \angle 120^\circ \text{ A}$ $I_A = I_{AB} - I_{CA} = 20.87 \angle -73.3^\circ \text{ A}$ $I_B = I_{BC} - I_{AB} = 13.95 \angle 173.4^\circ \text{ A}$ $I_C = I_{CA} - I_{BC} = 20.266.87 \text{ A}$ $P = V_{AB} \cdot I_{AB} \cdot \cos \phi_{AB} + V_{BC} \cdot I_{BC} \cdot \cos \phi_{BC} + V_{CA} \cdot I_{CA} \cdot \cos \phi_{CA}$ $= (240 \times 9.6 \times \cos 90^\circ) + (240 \times 16 \times \cos 30^\circ) + (240 \times 12 \times 1)$ $= \underline{\underline{6205.53 \text{ W}}}$
Q18	A 3-phase, 400V, 4-wire system has a star connected load with $Z_A = 20\Omega$, $Z_B = 15 + j10\Omega$ and $Z_C = j5\Omega$. Find the line currents, current through the neutral conductor and the total power consumed by the load.

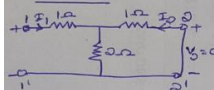
Ans	ABC sequence $V_A = 230.94 \angle 0^\circ \text{ V} \quad V_B = 230.94 \angle -120^\circ \text{ V} \quad V_C = 230.94 \angle 120^\circ \text{ V}$ $I_A = \frac{V_{AN}}{Z_A} = \frac{230.94 \angle 0^\circ}{40} = 11.54 \angle 0^\circ \text{ A}$ $I_B = \frac{V_{BN}}{Z_B} = \frac{230.94 \angle -120^\circ}{15 + j10} = 12.81 \angle -153.69^\circ \text{ A}$ $I_C = \frac{V_{CN}}{Z_C} = \frac{230.94 \angle 120^\circ}{j5} = 46.18 \angle 30^\circ \text{ A}$ $I_N = -(I_A + I_B + I_C) = 43.69 \angle -156.5^\circ \text{ A}$ $P = V_A I_A \cos \phi_A + V_B I_B \cos \phi_B + V_C I_C \cos \phi_C$ $= 230.9 \times 11.54 \times 1 + 230.9 \times 12.81 \times \cos 33.69^\circ + 230.9 \times 46.18 \times \cos 90^\circ$ $= \underline{5.128 \text{ kW}}$
Module 5	
Q19	a) A two port network 'A' has $[Z]_A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and another two port network 'B' has $[Y]_B = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$ If the two networks A and B are connected in series, find the Z parameters of the overall network. b) Find the equivalent π network of a two port network represented by the following equations. $V_1 = 2I_1 + I_2$ $V_2 = I_1 + 3I_2$

Ans	series-connection of two-port N/W a) $Z_A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ $[Y_B] = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$ $Z_B = [Y_B]^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ $Z_T = Z_A + Z_B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$ b) $V_1 = 2I_1 + I_2$ $V_2 = I_1 + 3I_2$ $Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ $Y = [Z]^{-1} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$ $Y_a = Y_{11} + Y_{21} = 0.4 \Omega^{-1}$ $Y_b = Y_{22} + Y_{12} = 0.2 \Omega^{-1}$ $Y_c = -Y_{12} = -Y_{21} = 0.2 \Omega^{-1}$ 
Q20	a) Find the h-parameters of the following two port network.  b) Derive the conditions of symmetry and reciprocity of a two port network in terms of ABCD parameters.
Ans	a)

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \times \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \times \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

Case-I ($V_2 = 0$)



$$h_{11} = \frac{V_1}{I_1} = \frac{I_1 [1 + (2 \parallel 1)]}{I_1}$$

$$= 1 + \frac{2 \times 1}{2+1} = 1.667$$

$$h_{21} = \frac{I_2}{I_1} = -I_1 \times \left(\frac{2}{2+1} \right) = -\frac{2}{3}$$

$$= -0.667$$

Case-II ($I_1 = 0$)



$$h_{12} = \frac{V_1}{V_2} = \frac{V_2 \times \left(\frac{2}{2+1} \right)}{V_2}$$

$$= \frac{2}{3} = 0.667$$

$$h_{22} = \frac{I_2}{V_2} = \frac{I_2}{V_2 \times (1 \parallel (1+2))} = \frac{1}{1 \parallel 3}$$

$$= \frac{1}{\left(\frac{1 \times 3}{1+3} \right)} = \frac{4}{3} = 1.333 \Omega^{-1}$$

b) $(V_1, I_1) = f(V_2, I_2)$

$V_1 = AV_2 - BI_2$ -----

$I_1 = CV_2 - DI_2$ -----

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$V_2 = 0,$

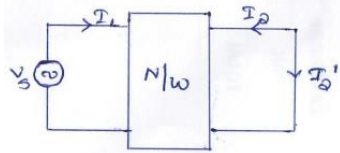
$B = \left. \frac{-V_1}{I_2} \right|_{V_2=0}$ = Shortcircuit Reverse transfer Impedance

$D = \left. \frac{-I_1}{I_2} \right|_{V_2=0}$ = Shortcircuit Reverse current gain

$A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$ = Open ckt, reverse Voltage gain

$C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$ = Open ckt reverse transfer admittance

Condition for Reciprocity



$$I_1 = \dots \quad I_2 = -I_2'$$

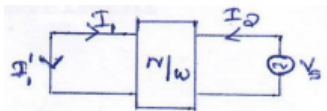
$$V_1 = V_S \quad V_2 = 0,$$

$$V_S = BI_2'$$

$$I_1 = DI_2'$$

$$V_S / I_2' = B$$

Case II



$$I_1 = -I_1' \quad I_2 = I_2$$

$$V_1 = 0 \quad V_2 = V_S$$

$$0 = AV_S - BI_2$$

$$AV_S = BI_2$$

$$I_2 = (AV_S)/B$$

$$-I_1' = CV_S - DI_2$$

$$-I_1' = CV_S - D(AV_S)/B$$

$$-I_1' = (BCV_S - DAV_S)/B$$

$$-I_1' = V_S ((BC-DA)/B)$$

$$\left| \begin{array}{l} \frac{V_S}{-I_1'} = \frac{1}{\frac{BC-DA}{B}} = \frac{B}{BC-DA} \\ \frac{V_S}{I_1'} = \frac{-B}{BC-DA} \end{array} \right|$$

For a network to be reciprocal,

$$\left. \frac{V_S}{I_1'} \right|_{V_2=0} = \left. \frac{V_S}{I_2'} \right|_{V_1=0}$$

$$\frac{-B}{BC-DA} = B$$

$$-1 = BC-DA$$

$$DA-BC = 1$$

		Condition for symmetry, $Z_{11} = \left. \frac{V_1}{I_1} \right _{I_2=0} = \left. \frac{AV_2 - BI_2}{CV_2 - DI_2} \right _{I_2=0} = \frac{A}{C}$ $Z_{22} = \left. \frac{V_2}{I_2} \right _{I_1=0} = \frac{D}{C} \quad \because I_1 = 0 = CV_2 - DI_2$ Condition for symmetry, $Z_{11} = Z_{22}$ ie. $\frac{A}{C} = \frac{D}{C} \quad \therefore A = D$